

Bilal Amin

+49 176 86534249 | write2bilalamin@gmail.com | [linkedin.com/in/aminbilal/](https://www.linkedin.com/in/aminbilal/) | Saarbrücken, Germany

EDUCATION

Universität des Saarlandes

Master's degree, Data Science and AI

Apr. 2024 – Present

Saarbrücken, Germany

- Relevant Coursework: Neural Networks: Theory and Implementation (NNTI), Elements of Machine Learning (EML), Machine Learning (ML), High-Level Computer Vision (HLCV), Generative AI, and Attacks Against ML Models (AAML).
- Erasmus+ Exchange Semester (Spring 2026) – University of Bergen, Norway: Focus on Deep Learning, Visual Data Science, and Reinforcement Learning.

National University of Sciences & Technology (NUST)

Bachelors in Mechatronics Engineering (Computer Science)

Sep. 2019 – Jun. 2023

Islamabad, Pakistan

- Thesis: Design and Development of Warehouse Management Robot (Fleet-Controlled Multi-Robot System).
- Final Grade: 1.7 (German grading system).

EXPERIENCE

Tutor – Human-Computer Interaction (HCI)

Universität des Saarlandes

Aug. 2024 – Present

Saarbrücken, Germany

- Guided undergraduate students in core HCI concepts, usability design, and interactive systems.
- Developed tutorial exercises and evaluated student assignments to reinforce learning objectives.
- Assisted the course instructor in running practical sessions and mentoring student projects.

Work Student – Internet Architecture Group

Max Planck Institute for Informatics

Oct. 2024 – Mar. 2025

Saarbrücken, Germany

- Developed automated frameworks in NS-3 for evaluating congestion control algorithms under varying network conditions.
- Created dashboards to visualize real-time performance metrics and conducted optimization analysis.

Data Intern

Convergent Business Technologies

Oct. 2023 – Mar. 2024

Islamabad, Pakistan

- Applied machine learning models for data-driven insights using Python and SQL.
- Created data pipelines, dashboards, and exploratory analysis for client datasets.

Web Development Intern

CureMD

Jun. 2023 – Aug. 2023

Lahore, Pakistan

- Designed and implemented database schemas, performed query optimization, and ensured seamless frontend-backend integration.
- Developed RESTful APIs in .NET (C#) and built interactive web components using HTML, CSS, and JavaScript.

PROJECTS

Warehouse Management Robot (Fleet Control System)

2023

- Developed an intelligent robotic system for warehouse automation capable of multi-robot coordination under a centralized fleet-control server.
- Implemented a rail-climbing mechanism enabling robots to retrieve payloads from multi-level storage racks with 60% faster traversal.
- Integrated suction-based pick-and-place system using camera vision and real-time task allocation algorithms.
- Achieved 40% improvement in load distribution and throughput, validated using CAD and ANSYS simulations.

Soloscale – Tech 4 Europe Hackathon (KIT Edition)

2025

- Developed an AI-driven platform that predicts customer demand and optimizes supply chain decisions using a Python-based backend and data pipelines.

Side Hustle Launchpad — Full-stack Web Application

2024

- Developed a full-stack web app to support small business creation through analytics dashboards and personalized recommendations using FastAPI and MongoDB.

Fine-Tuning and Data Selection Strategies for Chemical Language Models

2025

- Implemented LoRA, BitFit, and IA3 parameter-efficient fine-tuning on MoLFormer-XL for molecular property prediction.
- Applied influence functions for data selection, improving model generalization and reducing training set size.

TECHNICAL SKILLS AND INTERESTS

Languages: Python, C, C++, SQL, HTML/CSS, JavaScript

Frameworks / Libraries: FastAPI, Flask, TensorFlow, PyTorch, Keras, NumPy, Pandas, Matplotlib, Scikit-Learn, React, Node.js, Express.js

Tools & Platforms: Git, Linux, VS Code, PyCharm, Docker, Postman, MongoDB

Research Interests: AI Systems, Trustworthy AI, Generative AI, Deep Learning, Data-Centric AI, Fleet Control Systems

LEADERSHIP & VOLUNTEERING

Volunteer

Nov. 2021 – Apr. 2023

Kara Kamal Foundation

Islamabad, Pakistan

- Organized social drives, old-age home visits, and community fundraisers for families affected by famine and COVID-19.
- Led logistics and planning for volunteer initiatives improving access to healthcare and education.

Student Volunteer

Dec. 2019 – Oct. 2023

NUST Volunteer Club

Islamabad, Pakistan

- Coordinated environmental and humanitarian campaigns, including awareness drives and fundraising events.
- Supported cancer patients and low-income families through donation campaigns.

Technical Volunteer

Aug. 2021 – Oct. 2021

STEP (Space Technology Education and Popularization)

Islamabad, Pakistan

- Designed a curriculum introducing underprivileged children to robotics and programming using Scratch.
- Conducted workshops teaching practical tech skills and fostering STEM interest.

AWARDS & HONOURS

Runner-up, HeadStart Robowars (Sumo Category), 2019 – National Robotics Competition.

Best Design Award – Fruit Plucking Robot, NCRA Robotics Competition, 2022.

Finalist – Best Final Year Project (Warehouse Management Robot), NUST, 2023.

Selected Participant, Aspire Leadership Program for Aspiring New Leaders, 2023.

CONFERENCES & SEMINARS

International Conference on Robotics & Artificial Intelligence – Islamabad, 2019.

International Conference on Robotics & Artificial Intelligence – Islamabad, 2021.

Aspire Leadership Program – Global Online Leadership Seminar, 2023.